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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

MA2003-1 – VECTOR CALCULUS AND COMPLEX FUNCTIONS

(For CV, ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- In spherical coordinate system (r, θ, ϕ) , $r = a$, where a is a constant represents**
A) A plane
B) A Sphere with radius a
C) A cone
D) A cylinder with radius a
- A unit tangent vector to the surface $x = t, y = t^2, z = t^3$ at $t = 1$ is**
A) $\frac{\hat{i} + 2\hat{j} + 3\hat{k}}{\sqrt{14}}$
B) $\frac{\hat{i} - 2\hat{j} + 3\hat{k}}{\sqrt{14}}$
C) $\frac{2\hat{i} + \hat{j} + \hat{k}}{\sqrt{6}}$
D) $\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$
- $\text{div}(\text{curl } \vec{F}) =$ _____**
A) 1
B) 2
C) $\vec{0}$
D) 0
- $\vec{F} = yz\hat{i} + zx\hat{j} + xy\hat{k}$ is**
A) Irrotational vector field
B) Solenoidal vector field
C) Both solenoidal and irrotational vector field
D) $\nabla \cdot \vec{F} = 5$
- If $\vec{F}$ is conservative force field then $\text{curl } \vec{F}$ is _____**
A) $\nabla \cdot \vec{F}$
B) $\vec{F}$
C) $\vec{0}$
D) 0
- Using Green's theorem in the plane, the value of the integral $\frac{1}{2} \int_C xdy - ydx$ is**
A) Area of the region R bounded by C
B) 1
C) $1/2$
D) 0
- Gauss divergence theorem relates**
A) Line and surface integral
B) Surface and volume integral
C) Line and volume integral
D) Line integral, surface integral and volume integral
- The line integral $\int_C \vec{F} \cdot d\vec{r}$ represents:**
A) Area under the curve
B) Flux through the surface
C) Volume generated by rotation
D) Work done by the vector field $\vec{F}$ along path C
- If $f(z) = u(x, y) + iv(x, y)$ is analytic, then $f'(z) =$ _____**
A) $\frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}$
B) $\frac{\partial u}{\partial x} - i \frac{\partial v}{\partial x}$
C) $\frac{\partial v}{\partial y} - i \frac{\partial v}{\partial x}$
D) $\frac{\partial u}{\partial y} - i \frac{\partial v}{\partial x}$



10. If $u = 2x - x^2 + ay^2$ is to be harmonic, then a should be
 A) 0 B) 2
 C) 3 D) 1
11. If the real part of an analytic function is $u(x, y) = x^2 - y^2$, then using Milne-Thomson method, the analytic function $f(z)$ is:
 A) $f(z) = \bar{z}^2$ B) $f(z) = z^2$
 C) $f(z) = 2xy + i(x^2 - y^2)$ D) $f(z) = x^2 + y^2 + i 2xy$
12. The bilinear transformation $w = \frac{2z+3}{z+2}$ maps $z = -2$ to:
 A) $w = 0$ B) $w = -1$
 C) $w = \infty$ D) $w = 1$
13. The value of $\int_{|z|=2} \frac{dz}{z}$ is:
 A) 0 B) πi
 C) $-2\pi i$ D) $2\pi i$
14. Cauchy's integral formula states that if $f(z)$ is analytic inside and on a simple closed contour C and a is inside C , then:
 A) $f'(a) = \frac{1}{2\pi i} \int_C f(z) dz$ B) $f(a) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z-a} dz$
 C) $f(a) = \frac{1}{2\pi i} \int_C f(z) dz$ D) $f(a) = \int_C f(z) dz$
15. If $f(z) = \frac{10+6z^2}{(z-1)(z+2)(z-3)^3}$ then $z = 3$ is a pole of $f(z)$ of order _____
 A) 3 B) 1
 C) 2 D) 5
16. A Laurent series of $f(z)$ around $z = z_0$ can be written as:
 A) $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ B) $\sum_{n=0}^{\infty} a_n e^{n(z-z_0)}$
 C) $\sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$ D) $\sum a_n \cos n(z - z_0)$
17. If $f(x)$ is even, its Fourier series contains:
 A) Only sine terms B) Constant only
 C) Both sine and cosine terms D) Only cosine terms
18. The Fourier series of $f(x) = x$ on $[-\pi, \pi]$ contains:
 A) Only sine terms B) Only cosine terms
 C) Both sine and cosine terms D) Constant term only
19. If $F\{f(t)\} = F(\omega)$, then $f(t - a) =$ _____
 A) $F(\omega)e^{-i\omega a}$ B) $F(\omega)e^{i\omega a}$
 C) $F(\omega - a)$ D) $F(\omega + a)$
20. Fourier transform of $f(t) = e^{-|t|}$ is:
 A) $\frac{1}{1 + \omega^2}$ B) $\frac{2}{1 + \omega^2}$
 C) $\frac{1}{\omega}$ D) $\frac{1}{\omega^2}$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I		Marks	BT*	CO*	PO*
1.	a) Determine the magnitude of velocity and acceleration at $t = 0$ of a particle moving along a curve whose parametric equations are $x = e^{-t}$, $y = 2 \cos 3t$, $z = 2 \sin 3t$, where t is the time.	6	L*3	1	2
	b) Prove that cylindrical coordinate system is orthogonal curvilinear coordinate system.	6	L2	1	1
	c) Determine whether the following force field is conservative or not. $\vec{F} = 2xz \hat{i} + (x^2 - y) \hat{j} + (2z - x^2) \hat{k}$	4	L2	2	1

2. a) Find the total work done in moving a particle in a force field given by $\vec{F} = 3xy\hat{i} - 5z\hat{j} + 10x\hat{k}$ along the curve $x = t^2 + 1, y = 2t^2, z = t^3$ from $t = 1$ to $t = 2$. 6 L3 2 2
 b) State and prove the Green's theorem. 6 L2 2 1
 c) What is the greatest rate of increase of $u = x^2 + yz^2$ at the point $(1, -1, 3)$? 4 L2 1 1
3. a) Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at $(1, 2, 2)$. 6 L2 1 1
 b) Using Stoke's theorem evaluate $\int_C \vec{F} \cdot d\vec{r}$ for the vector field $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ round the rectangle in the plane $z = 0$ and bounded by the lines $x = 0, y = 0, x = a$ and $y = b$. 6 L3 2 2
 c) Prove that $\nabla \cdot (\nabla \times \vec{F}) = 0$. 4 L2 1 1

Unit – II

4. a) Show that $v = e^x \sin y$ is harmonic. Find u such that $f(z) = u + iv$ is analytic. 6 L2 3 1
 b) If $f(z) = u(x, y) + i v(x, y)$ is analytic in a region R , then prove that $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ & $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$. 6 L2 3 1
 c) Evaluate $\int_C \frac{dz}{z^2 - z}$, if $C: |z| = \frac{1}{2}$ using the Cauchy's integral formula. 4 L3 4 2
5. a) Evaluate $\int_C \frac{z^2}{(z-1)^2(z+2)} dz$ where C is the circle $|z| = \frac{5}{2}$ using the residue theorem. 6 L3 4 2
 b) Write the Laurent's series expansion of $\frac{1}{z^2 - 3z + 2}$ in the region $|z| > 2$. 6 L2 4 1
 c) Show that $w = \log z$ is analytic. 4 L2 3 1
6. a) Evaluate the bilinear transformation which maps $z = 0, 1, \infty$ onto $w = -1, -i, 1$. 6 L3 3 1
 b) State and prove the Cauchy's integral theorem. 6 L2 4 1
 c) Find the poles and residues at each pole of the function $f(z) = \frac{2z+1}{z^2 - z - 2}$. 4 L2 4 2

Unit – III

7. a) Evaluate a_0, a_1, b_1 of the Fourier series of the following data.

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
y	1	1.4	1.9	1.7	1.5	1.2	1

- b) Solve the Fourier series of $f(x) = x^2$ in $(-\pi, \pi)$ with $f(x + 2\pi) = f(x)$. 6 L3 5 1
 c) Solve the half range Fourier sine series of $f(x) = x$ in $0 < x < \pi$. 6 L2 5 2
 4 L2 5 1
8. a) State and prove the convolution theorem of Fourier transforms. 6 L2 5 1
 b) Evaluate the Fourier transform of $e^{-a|t|}$, $t \geq 0, a > 0$. 6 L3 5 2
 c) Show that $F_c\{t f(t)\} = \frac{d}{d\omega}\{F_s(\omega)\}$. 4 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
